



Aetherum

— PRODUCT OVERVIEW

The Crypto Lending Platform That *Explains Itself*.

Every other platform is building faster pipes. Aetherum gives your members a lending decision they can actually understand — and that explanation is the moat.

12

DACS RISK
PILLARS

20

PATENTS PENDING

5,000+

CREDIT UNIONS
REACHABLE

SCHEDULE A DEMO

↓ DOWNLOAD PDF



Built for the member. *Deployed for the credit union.*

The market has been focused on speed for the institution. Aetherum is focused on trust for the member — and hands the credit union the infrastructure to deliver it.

LENDING

Crypto-Backed Lending

Members keep their crypto exposure, avoid a taxable event, and get USD liquidity — all through their trusted credit union. Aetherum handles the infrastructure. You own the relationship.

UNDERWRITING

Smart Review

AI-powered underwriting guidance evaluates the full borrowing position and explains it in plain language — recommended amount, risk level, and reasoning, instantly, at the point of decision.

PATENT PENDING

TAX INTELLIGENCE

Tax-Position Score & Optimizer (TPSOE™)

Scores the tax efficiency of each collateral position and ranks pledge assets to minimize member tax liability — a feature no D2C competitor offers institutions.

PATENT PENDING

DEBT INSTRUMENTS

CU Debt Instrument — Audit-Native Tokenized Loan Participation

Aetherum's CU Debt Instrument module tokenizes pools of member loans as ERC-3643 debt tokens with three properties unique to regulated depositories: transfer-layer gating on regulatory counterparty eligibility (the "eligible organization" test as a condition of settlement), policy-gated default routing missed coupons and LTV breaches through TxGuard's pre-signing policy engine before liquidation, and examiner-native audit emission where every lifecycle event becomes a compliance receipt.

AETHERUM-021 · PATENT PENDING



A *risk assessment model* built for crypto collateral.

DACS quantifies a borrower's position beyond traditional metrics — in language the member can actually understand. Twelve pillars of crypto risk assessment, backed by 21 patent-pending innovations, with a plain-language explanation behind every decision. Not a number on a screen; the reasoning behind it.

SEE THE 12 PILLARS →



— 02 · DACS — THE TWELVE PILLARS

- | | | | |
|--|--|--|--|
| 1 Wallet Age
on-chain longevity | 2 Diversification
portfolio spread | 3 On-Chain Activity
active management | 4 Volatility
collateral resilience |
| 5 Tax-Position Score
TPSOE · patent pending | 6 TAM Bonus
verified holdings | 7 On-Chain Identity
CUMember · patent pending | 8 Privacy Computation
FHE-DACS · patent pending |
| 9 Dynamic LTV
DACS-LTV · patent pending | 10 Ledger Collateral Control
patent pending | 11 XLS-66 LoanBroker
patent pending | 12 Key Governance & Audit
patent pending |

Tier 1 members access up to 70% LTV, Tier 2 up to 65%, Tier 3 up to 55% — enforced at the blockchain layer, no manual override.



Compliance *built in*, not bolted on.

IDENTITY

On-Chain Member Identity

ERC-3643 + ERC-7943 compliant verification. Each member gets a portable ONCHAINID attestation — KYC verified, jurisdiction gated, collateral-scored on-chain. Built on the T-REX standard for regulated RWAs.

PATENT PENDING

PRIVACY

Privacy-Preserving Underwriting (FHE-DACS)

Encrypted risk assessment via homomorphic encryption — member financial data is never exposed in plaintext during underwriting.

PATENT PENDING

ENFORCEMENT

ERC-7943 Collateral Token

The official Ethereum Universal RWA Interface — on-chain eligibility checks, regulatory freeze, and forced transfer, all delegating through three ERC-3643 compliance modules.

PATENT PENDING

CU OWNERSHIP

CU-Owned Identity Infrastructure

Aetherum deploys a ClaimIssuer and IdFactory contract on Base mainnet for each credit union — owned by the CU, not Aetherum. Member KYC eligibility is stored as a cryptographic claim on the member's ONCHAINID. Every claim issuance, transfer, and revocation generates an audit receipt for NCUA examination. Aetherum's signing authority is purpose-3 only and revocable by the CU at any time.

ERC-734 / ERC-735 · Base Mainnet

ERC-734 / ERC-735

ONCHAINID

Per-credit-union identity infrastructure deployed on Base mainnet. Each member's KYC clearance is stored as a cryptographic claim on their ONCHAINID — owned by the credit union, enforced by the chain. Every claim issuance and revocation generates a tamper-evident compliance receipt for NCUA examination.

CU-OWNED CONTRACTS

PATENTS PENDING



Every decision *examiner-ready* from day one.

RECEIPTS

On-Chain Compliance Receipts

SHA-256 hash of every decision written to Ethereum. Examiners receive a verification URL per decision; sensitive data stays off-chain. No other lending infrastructure provides this.

EXPLAINABILITY

DACS Explainability Layer

Auto-generated record after every scoring run — factor breakdowns, risk tier, peer comparison, plain-language rationale, and 12 CFR Part 723 citation. PDF export for NCUA submission.

PATENT PENDING

SUPERVISORY

Supervisory Control & Audit

Documented human-in-the-loop control over every position — NCUA-aligned event types, officer attribution, on-chain attestation, CSV examination export with Etherscan links.

PATENT PENDING

DILIGENCE

DDQ Engine

387-entry security knowledge base spanning NCUA guidance, custody, BSA/AML, and the full SIG framework. A CU IT team can complete a full vendor security review without a call.

LIVE

TRANSACTION SECURITY

AetherumTxGuard — Pre-Signing Transaction Verification

Every transaction decoded, policy-checked, and logged before it reaches a signing key — per-institution policy enforcement with audit-native compliance logging for NCUA-regulated credit union lending infrastructure.

PATENT PENDING



Money moves. *Collateral is watched.*

DISBURSEMENT

Multi-Rail Stablecoin

Pluggable multi-rail disbursement — USD & SGD across EVM networks, cross-chain transfer, and programmable MPC wallets. Migrating to Arc — The Economic OS · Stablecoin-native L1 — for USDC-native gas, sub-second deterministic finality, and native Circle-stack integration (USDC, CCTP, Gateway). Mainnet Q3 2026.

LIVE

CARD ISSUANCE

Card Funding Bridge

On approval, triggers white-label virtual card issuance funded by USDC disbursement. Member receives a branded credit union card instantly. Completing the borrow→disburse→spend loop for credit union members.

PATENT PENDING

SURVEILLANCE

Collateral Intelligence Engine (CIE™)

Every position monitored 24/7 — finality-aware on-chain event processing, live LTV, DACS updated on every price move, institution-authorized liquidation control. Live staking yield feed for SOL collateral: $APY = \frac{\text{getInflationRate}}{\text{staking ratio}}$, cached 2 hours, surfaced on the member dashboard.

PATENT PENDING

MONITORING

Native LTV Monitoring

BTC and ETH collateral polled every 60 seconds via Blockstream and Alchemy — no third-party dependency. 30-day rolling volatility feeds DACS directly.

LIVE

OVERSIGHT

Liquidation Escalation

Three-tier human oversight: Tier 1 auto-executes, Tier 2 pages your risk officer with a 30-min window, Tier 3 requires officer authorization for \$250K+ or 40%+ drops. Every action timestamped.

LIVE



— 06 · WHO RUNS IT

A view *for every seat* at the credit union.

LENDING

Loan Officer Portal

Review applications, assess DACS scores, manage active loans, and action margin-call alerts — purpose-built for crypto-collateralized lending.

COMPLIANCE

Compliance Officer

Siloed KYC/AML queue, sanctions monitoring, and audit-ready reporting — full visibility and control without touching operational loan data.

REGULATORY

NCUA Examiner View

Read-only audit dashboard scoped per CU. Live positions, LTV ratios, liquidation events, full BSA/AML trail. Zero custody access, zero execution permissions.

MEMBERS

Member Crypto Insights

Consent-based visibility into the assets members already own — DACS eligibility scoring and an opt-in loan pipeline. See what's invisible today.

PATENT PENDING



Four steps. *You own the relationship.*

1

Member deposits crypto collateral into an institutional custody wallet — held directly by the credit union, never by Aetherum.

2

DACS scores the borrower and Smart Review AI evaluates the application against NCUA guidance — returning a structured decision with compliance notes and a full audit trail.

3

The credit union issues the loan using existing infrastructure. Aetherum handles the rest.

4

Automated monitoring manages margin calls, top-ups, and liquidation if needed — with human oversight on every large position.

GENIUS Act ready. Aetherum was built inside the NCUA's regulatory perimeter — on-chain identity, per-CU custody, and compliance-first infrastructure designed to operate within supervision, not around it.

Full borrow → disburse → spend loop live in sandbox: Mesh collateral intake, RLUSD disbursement on Base, Nium card spend via Visa.



LIVE INTEGRATIONS

Jack Henry FIN
5,000+ CUs
reachable

**Corelation
KeyStone**
~290 CUs day one

**Fiserv
AppMarket**
DNA platform

**FIS Code
Connect**
IBS Open APIs

XRPL / XLS-66
RLUSD loan
currency

**Pharos · Circle
CCTP**
USDC, live testnet

Mesh
Collateral Transfer
Rail

Plaid
Bank Data
Connectivity ·
Certified

**Arc Transaction
Memos**
On-chain audit trail
attached to every
disbursement. Loan
ID, member
identifier, and
branch code travel
with the USDC
transfer, indexed
on-chain, readable
by an NCUA
examiner without
querying a vendor
database.

Sardine
COMPLIANCE &
FRAUD
INFRASTRUCTURE

REVENUE MODEL

- 01 Platform licensing fee (annual SaaS)
- 02 Per-loan origination fee
- 03 DACS API access for third-party integrations
- 04 RWA tokenization service fees (coming soon)
- 05 Staking yield fee — 15bps on total staking yield via YieldGuardService

Credit unions keep 100% of loan interest income.



Aetherum

— 09 · GET IN TOUCH

Bring crypto-backed lending *to your credit union.*

Schedule a 30-minute demo.

We'll walk through the platform live — DACS, Smart Review, the audit-native trail — and answer any diligence questions.

SCHEDULE A DEMO

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